

CLASSES

Task

- *Homework*
- *Chore*
- *Activity*
- *Lecture*
- *Work*

TimeSlot (every 15 min)

BlockedWindow

- *AcademicBreak*
- *Sleep*

PROPERTIES

INPUT FACTS

hasDeadline

hasDuration

startTime

endTime

startDate

endDate

hasEnergyLevel

- *Low = 1*
- *Med = 2*
- *High = 3*

requiresEnergyLevel

- *Low = 1*
- *Med = 2*
- *High = 3*

frequency (None/Daily/Weekly)

priorityLevel

isFixed

INFERRED FACTS

scheduledStatus

urgencyLevel

conflictsWith

AXIOMS (Rules)

URGENCY

If a task has a deadline and it's due in less than 24 hours, its urgency is High.

If a task has a deadline between 24 hours and a week away, its urgency is Medium.

If a task has a deadline more than a week away, its urgency is Low.

If a task has no deadline at all, it defaults to Low urgency.

ENERGY

If a task requires more energy than a given time slot has, the task can't be scheduled during that time.

BLOCKED TIME

Nothing recurring gets scheduled during academic breaks. One-off tasks specifically asked to schedule during a break are allowed.

Nothing gets scheduled during sleep hours.

FIXED TASKS

Lectures never get rescheduled.

Tasks marked as fixed never get rescheduled, regardless of category.

CONSIDERED SLOTS

Homework can only be scheduled in time slots that end before the homework's deadline.

(Nothing gets scheduled past the due date.)

Activities do not get scheduled before 4pm on weekdays.

Nothing gets scheduled with less than 15 min of breathing room before/after another task. This rule doesn't apply to lectures, since those can't be rescheduled.

CONFLICTS

Two tasks scheduled in overlapping time slots are marked as conflicting.

Conflicting tasks are resolved by:

1. If only one is fixed, the non-fixed one gets moved.
2. Otherwise the one with lower urgency gets moved.
3. If urgency is tied, the lower priority one gets moved.
4. If that is tied, the one with the later deadline gets moved.
5. If they're still tied but different task types, the lowest one in this hierarchy gets moved:
Activity → Chore → Work → Homework → Lecture
6. If they're the same type, randomly choose one to be moved.

<https://inferensys.com/glossary/enterprise-knowledge-graphs/knowledge-representation-languages/owl-ontology>